

i, j, k, l, m, n	
x	
$t, \text{ term}$	$::=$ $ \quad x$ $ \quad \mathbf{letrec\ } lb_1 \mathbf{\ and \ ... \ and \ } lb_n \mathbf{\ in \ } t$
$lb, \text{ letrec_binding}$	$::=$ $ \quad x = t$
$b, \text{ binding}$	$::=$ $ \quad x$
$E, \text{ environment}$	$::=$ $ \quad b_1, \dots, b_n$ $ \quad \mathbf{nil} \quad \quad \quad \mathbf{M}$ $ \quad E_1, \dots, E_n \quad \quad \quad \mathbf{M}$
formula	$::=$ $ \quad \text{judgement}$
Typing	$::=$ $ \quad E - \mathbf{ok}$ $ \quad E - t$ $ \quad E - lb \mathbf{\ gives \ } E'$
judgement	$::=$ $ \quad \text{Typing}$
user_syntax	$::=$ $ \quad i$ $ \quad x$ $ \quad t$ $ \quad lb$ $ \quad b$ $ \quad E$

$E| - \mathbf{ok}$

$\frac{}{\mathbf{nil}| - \mathbf{ok}} \quad \text{TYPING_OK_NIL}$

$\frac{E| - \mathbf{ok}}{E, x| - \mathbf{ok}} \quad \text{TYPING_OK_X}$

$E| - t$

$\frac{E_1, x, E_2| - \mathbf{ok}}{E_1, x, E_2| - x} \quad \text{TYPING_TERM_X}$

<<no parses (char 35): $E, E_1, \dots, E_n | - lb_1 \text{ gives } E_1 \text{ .***. } E, E_1, \dots, E_n | - lb_n$ >>

$E, E_1, \dots, E_n| - t$

$E| - \mathbf{letrec\ } lb_1 \mathbf{\ and \ ... \ and \ } lb_n \mathbf{\ in \ } t$

$$\boxed{E| - lb \textbf{gives} E'}$$

$$\frac{E| - t}{E| - x = t \textbf{gives} x} \quad \text{TYPING_LB_SIMPLE}$$

Definition rules: 4 good 1 bad

Definition rule clauses: 9 good 1 bad